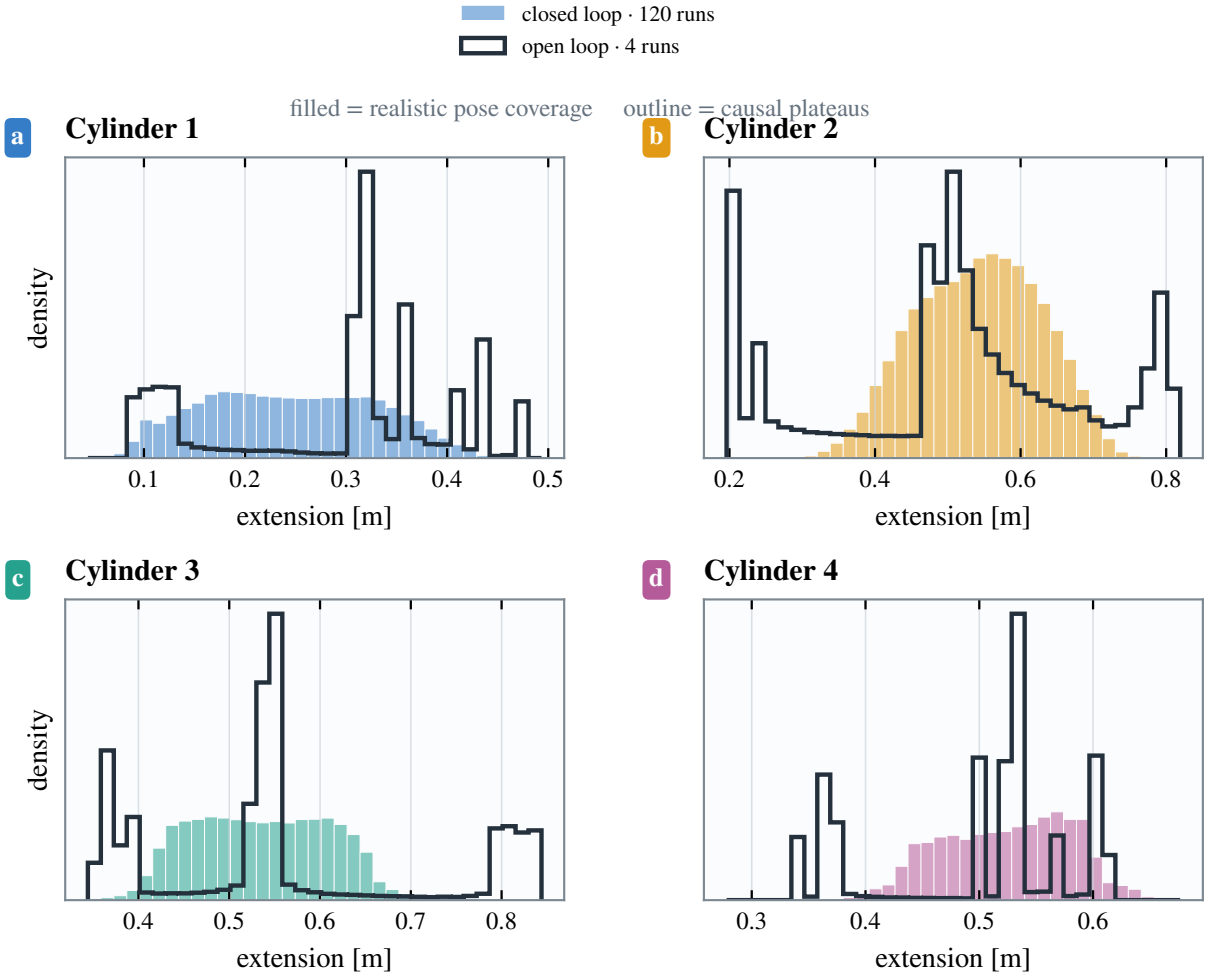


Figure 4.4: Extension coverage of the two campaigns, per cylinder and pooled over runs (closed-loop filled, open-loop outline). The closed-loop data fills the reachable range smoothly the open-loop data concentrates into the plateaus where the valve is identifiable.

and the join is accepted only when the sixteen-dimensional cylinder state agrees between them sample for sample. The resulting command-conditioned dataset contains 452 400 transitions at 0.05 s drawn from 1 040 aligned runs. The logged valve command is carried through this join, but only as a supervision target for the model's internal valve head. It is never a runtime input, which is the property that lets the ROM be driven by desired cylinder motion alone (Section 5.4).

Two closed-loop bodies of Simscape material therefore exist, they are easy to confuse, and only one of them trains anything in this thesis. The distinction is worth making sharply.

The *aligned export*[1] is the one that trains the inverse, forward and reduced-order models. It is a nominal-plant closed-loop collection of 1 040 runs yielding 452 400 aligned transitions at 0.05 s, split by complete run into 936 training, 52 validation and 52 test. It predates this thesis: it was inherited as a frozen export, and its collection script is not part of this work, so what can be stated about it is what can be measured from it, which is the run count, the sample time, the channel set, the split, and the audit results reported in Section 4.10. That is a reproducibility limitation and is recorded as one. Every property